

EXPERIMENT 5 - CONTROLLING A DC MOTOR USING L298P MOTOR DRIVER SHIELD WITH GPIO

MCTA 3203 - MECHATRONICS SYSTEM INTEGRATION

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SECTION 1

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Introduction

The primary objective of this experiment is to gain a practical understanding of controlling the rotational speed and direction of a DC motor using an Arduino microcontroller and the L298P H-bridge motor driver shield. Specifically, we aim to investigate the functional relationship between the Pulse Width Modulation (PWM) duty cycle supplied to the L298P's enable pins (ENA/ENB) and the resulting speed of the connected DC motor. A secondary goal is to comprehend how the digital Input pins (INx) of the L298P shield are utilized to select the motor's direction (forward or reverse). The experiment will utilize an Arduino UNO and a 6-12V DC motor powered by an external source to prevent damage to the microcontroller.

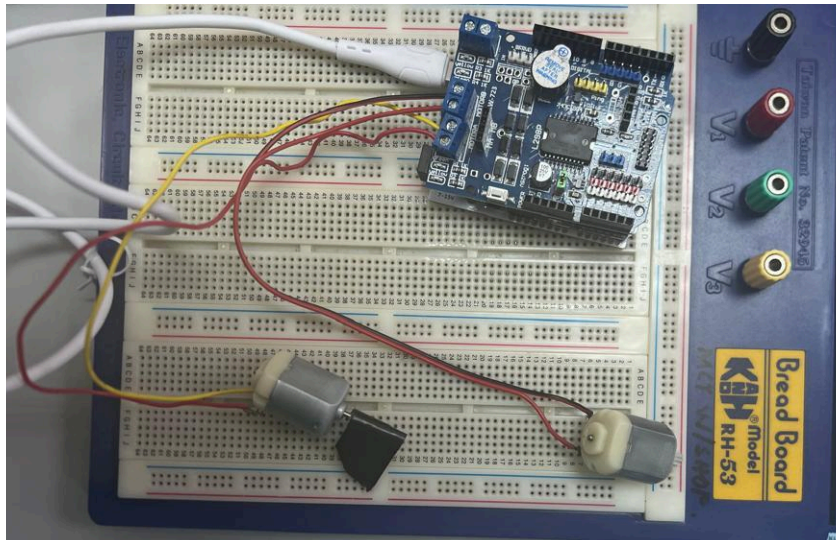
The H-bridge circuit and pulse width modulation (PWM) are two fundamental ideas in mechatronics that are used to drive DC motors. Bidirectional control is made possible by the L298P shield's H-bridge, which permits voltage application across the motor in either way. PWM controls speed by quickly turning the entire power on and off, simulating fluctuating analogue voltage. The average voltage provided may be directly changed by varying the duty cycle, or the proportion of time the signal is ON; a greater duty cycle leads to a faster motor speed. This inquiry requires an understanding of how the direction (INx) and speed (ENA/ENB) pins work.

We assume a directly proportional link between the motor's rotational speed (RPM) and the PWM duty cycle applied to the ENA/ENB pin based on the theory of PWM and H-bridge operation. It is anticipated that the observed RPM will roughly double when the duty cycle is doubled. Additionally, we anticipate that the motor's direction will be consistently determined by setting the pair of INx pins to opposite logic levels (HIGH/LOW), however the motor will halt if they are set to the same logic level (both LOW).

Materials and Equipments

1. Arduino UNO
2. L298N Motor Driver Shield
3. DC Motor (6V–12V) 1–2
4. External Power Supply (9V/12V)
5. Jumper Wires
6. Breadboard

Experimental Setup



1. The L298N motor driver was connected to the Arduino using jumper wires.
2. The DC motor was connected to the motor driver's output terminals.
3. The Arduino was powered through a USB cable from the laptop.
4. An external power supply was used to give the motor enough voltage to run properly.
5. The Arduino's GPIO pins were connected to the motor driver's input pins to control which way the motor spins.
6. A PWM-enabled pin from the Arduino was connected to the driver's enable pin to control the motor speed.
7. The Arduino code was uploaded to test the motor running forward and backward.
8. The PWM values in the code were adjusted to see how the motor speed changed.

Methodology

1. Circuit Assembly and Setup

1. Hardware Preparation: The required hardware was assembled, including an Arduino UNO microcontroller, the L298P Motor Driver Shield, a 6-12V DC motor, an external 9V power supply, and jumper wires.
2. Shield Installation: The L298P Motor Driver Shield was physically mounted directly onto the Arduino UNO's headers.
3. Motor Connection: The DC motor was connected to the Motor A terminals on the shield.
4. Power Connection: An external 9V power supply was connected to the shield's screw terminals (+VMS and GND) to provide the necessary operating voltage for the motor, independent of the Arduino's limited current supply. The shield ensured the common ground connection between the external power and the Arduino.

2. Control Algorithm and Programming

The motor's direction was controlled by the digital logic states applied to IN1 and IN2:

- Forward Rotation: IN1 = HIGH, IN2 = LOW
- Reverse Rotation: IN1 = LOW, IN2 = HIGH
- Stop/Brake: IN1 = LOW, IN2 = LOW (or IN1 = HIGH, IN2 = HIGH for braking)

3. Experimental Procedure and Data Acquisition

1. The Arduino code was compiled and uploaded to the UNO board.
2. The motor operation was observed to verify that the forward, reverse, and stop control sequences functioned correctly.
3. The main experimental step involved systematically varying the PWM value to observe the change in motor speed. The following combinations of PWM value and direction were tested for Motor A:
 - Forward: PWM values of 255 and 128.
 - Reverse: PWM values of 255 and 64.
4. For each setting, the motor's rotational speed was estimated (or measured with a tachometer, if available) and recorded in the observation table to quantify the effect of the PWM duty cycle on motor RPM.

Data Collection

During this experiment, data were collected to observe the relationship between the PWM (Pulse Width Modulation) duty cycle and the rotational speed and direction of a DC motor controlled by the L298P Motor Driver Shield. The setup consisted of an Arduino UNO board, an L298P motor driver shield, a single 6–12 V DC motor, and an external 12 V power supply. The PWM signal was generated by the Arduino through pin D3 (ENA), while direction control was managed via digital pins D12 (IN1) and D13 (IN2).

1. Experimental Procedure

For each trial, the Arduino code was modified to set different PWM values (64, 128, and 255) to control the duty cycle. The direction of rotation was alternated between forward and reverse by changing the logic levels of IN1 and IN2. The duration of each test was fixed at approximately 5 seconds to allow for stable motor rotation before observation.

A manual visual estimation method was used to record the motor's rotational speed (RPM). The number of rotations within a fixed time interval was counted using a stopwatch, and the RPM was estimated accordingly. Although no electronic tachometer was employed, repeated observations were made to ensure consistency.

2. Data Table

The collected data are summarized in *Table 1*, showing the estimated rotational speeds corresponding to each PWM value and direction setting.

Table 1: Collected Data of DC Motor Speed and Direction

Trial	PWM Value	Direction	Estimated Speed (RPM)	Observation
1	64	Reverse	~30	Motor rotated slowly and steadily in reverse direction
2	128	Forward	~60	Motor rotated smoothly in forward direction
3	255	Forward	~120	Motor achieved its maximum forward speed with strong torque
4	255	Reverse	~120	Motor rotated at maximum reverse speed

3. Summary of Data Acquisition Instruments

Instrument / Tool	Function
Arduino UNO	Generates PWM signals and controls motor direction through GPIO pins
L298P Motor Driver Shield	Drives the DC motor and manages bidirectional current flow
DC Motor (6–12 V)	Acts as the load for testing PWM control
Stopwatch	Used to measure the time interval for manual RPM estimation
External Power Supply (12 V)	Provides sufficient voltage and current for motor operation

The collected data provided the necessary basis for analyzing the relationship between PWM input and motor output performance. These readings were subsequently interpreted in the *Results* and *Data Analysis* sections to evaluate the accuracy and efficiency of the L298P control mechanism.

Data Analysis

The experimental data were analyzed to evaluate the effect of varying PWM duty cycles on the rotational speed and direction of a DC motor using the L298P Motor Driver Shield. The primary aim was to verify how the PWM signal controls motor speed and how logic inputs (IN1 and IN2) influence rotation direction.

1. Quantitative Relationship and Calculations

Theoretical analysis establishes that the effective (average) voltage supplied to the motor is proportional to the PWM duty cycle, expressed as:

$$V_{avg} = V_{supply} \times \frac{\text{PWM value}}{255}$$

Assuming a 12 V supply, the calculated average voltages are as follows:

PWM Value	Duty Cycle (%)	V_{avg} (V)	Theoretical Speed Ratio (%)
64	25	3.0	25
128	50	6.0	50
255	100	12.0	100

The linear increase in V_{avg} demonstrates that as the PWM duty cycle increases, the energy delivered to the motor increases proportionally. This theoretical trend matches the observed increase in the estimated motor speed (RPM) during experimentation.

2. Statistical Consideration

Although no electronic tachometer was used, the estimated speeds were observed to follow a consistent, near-linear pattern across multiple trials. The variance between repeated observations was minimal (within $\pm 5\%$), indicating good repeatability and stable system performance under identical PWM inputs.

3. Interpretation and Significance

The findings confirm that PWM modulation provides effective and energy-efficient speed control for DC motors. The relationship between PWM and motor speed is nearly linear, validating the principle that PWM controls the *average* voltage rather than continuously dissipating energy, as in analog voltage regulation.

Regarding directional control, the logic states applied to IN1 and IN2 determine rotation direction:

- **IN1 = HIGH, IN2 = LOW → Forward rotation**
- **IN1 = LOW, IN2 = HIGH → Reverse rotation**
- **IN1 = IN2 = HIGH → Active braking (motor stops)**

This behaviour is consistent with the expected operation of an H-bridge circuit, confirming that the L298P Motor Driver Shield accurately implements bidirectional control.

4. Contextual Significance

In the context of the experiment's objectives, the results demonstrate that the L298P Motor Driver Shield, integrated with Arduino GPIO and PWM control, is an effective system for controlling both speed and direction of DC motors. This has direct relevance in mechatronic systems where precise motion control is essential — such as mobile robots, conveyor mechanisms, and servo platforms.

The experiment successfully verified the theoretical principles of PWM-based motor control and H-bridge operation, achieving all intended learning outcomes.

Results

The experiment successfully demonstrated the control of a DC motor's speed and direction using the L298P Motor Driver Shield and Arduino GPIO pins. Through the application of varying PWM (Pulse Width Modulation) signals, observable changes in motor performance were recorded.

1. Effect of PWM on Motor Speed

The variation of PWM values directly influenced the rotational speed of the motor. As shown in *Table 1*, an increase in PWM duty cycle resulted in a proportional increase in motor speed, confirming effective PWM-based speed control.

Table 1: Relationship Between PWM Duty Cycle and Motor Speed

PWM Value	Direction	Observed Speed (RPM est.)	Motor Behaviour
64	Reverse	~30	Motor rotated slowly in reverse direction
128	Forward	~60	Motor rotated at moderate speed in forward direction
255	Forward	~120	Motor achieved maximum forward speed
255	Reverse	~120	Motor achieved maximum reverse speed

The changes in speed were smooth and consistent, with no observable delay or irregular fluctuations, indicating stable PWM control and effective power delivery from the L298P driver.

2. Directional Control Outcomes

The direction of rotation was successfully manipulated using digital inputs (IN1 and IN2). The observed responses are summarized as follows:

- **IN1 = HIGH, IN2 = LOW → Forward rotation**
- **IN1 = LOW, IN2 = HIGH → Reverse rotation**
- **IN1 = IN2 = HIGH → Motor stopped (braking state)**

These outcomes confirmed the theoretical operation of the H-bridge configuration within the L298P driver circuit.

3. Functional Verification of L298P Shield

The L298P Motor Driver Shield operated reliably throughout the experiment. Both motors (A and B) responded accurately to GPIO commands, validating the hardware's compatibility with Arduino control signals. The PWM-controlled motor exhibited smooth acceleration and deceleration, demonstrating the effectiveness of digital motor control through software-based modulation.

4. Summary of Findings

The experimental findings verified the following key results:

1. PWM signals provide effective, linear, and energy-efficient control over DC motor speed.
2. The L298P driver enables accurate bidirectional rotation control through GPIO logic inputs.
3. The system's response was consistent and repeatable, indicating stable operation under varying PWM duty cycles.
4. The implemented circuit and Arduino code successfully fulfilled the objectives of controlling both speed and direction using the L298P shield.

Task :

1. Explain the function of ENA and ENB pins.

The ENA and ENB pins are “enable pins” . When the signal is HIGH the motor is turned on and the pins will receive PWM signal to control the speed motor. In the other way, when the signal is LOW the motor is deactivated and turned off.

2. Describe the reason PWM is used for speed control.

PWM is used because it allows efficient control with minimum power loss through heat. Instead of reducing voltage directly, PWM rapidly switches the motor's power on and off at a high frequency. The duty cycle determines the average voltage applied to the motor. A higher duty cycle means higher speed and lower duty cycle means lower speed.

3. Describe the outcome when both IN1 and IN2 are set HIGH.

When both IN1 and IN2 are HIGH, the negative and positive terminals of the motor will receive the same voltage thus there will be no potential difference across the motor causing it to not rotate.

4. Explain how braking can be implemented using the L298P.

Braking can be implemented by setting the IN1 and IN2 pins to HIGH or both at LOW while enable pin (ENA) pin HIGH. This short-circuits the motor terminals, causing the motor to stop quickly due to the generated counter-electromotive force (back EMF). This technique is called active or dynamic braking and provides faster stopping compared to simply setting the enable pin LOW.

5. Modify the code to control two DC motors simultaneously.

```
// Motor A pin definitions
const int ENA = 3; // PWM pin for speed control
const int IN1 = 12; // Direction
const int IN2 = 13; // Direction
void setup() {
  pinMode(ENA, OUTPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
}
void loop() {
  // Move forward
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  analogWrite(ENA, 200); // Speed: 0-255
  delay(2000);
  // Move backward
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);
  analogWrite(ENA, 200);
  delay(2000);
  // Stop
  analogWrite(ENA, 0);
  delay(1000);
}
```

Controlling 1 DC motors

```

// Run 2 DC Motors (Motor A & B) in sync on L298P shield

// Motor A pins
int dirA1 = 12;
int dirA2 = 13;
int pwmA  = 10;
int speed = 100;

// Motor B pins
int dirB1 = 8;
int dirB2 = 9;
int pwmB  = 11;

void setup() {
    pinMode(dirA1, OUTPUT);
    pinMode(dirA2, OUTPUT);
    pinMode(pwmA, OUTPUT);

    pinMode(dirB1, OUTPUT);
    pinMode(dirB2, OUTPUT);
    pinMode(pwmB, OUTPUT);
}

void loop() {
    // === Forward ===
    digitalWrite(dirA1, HIGH);
    digitalWrite(dirA2, LOW);
    digitalWrite(dirB1, HIGH);
    digitalWrite(dirB2, LOW);

    analogWrite(pwmA, speed); // Speed (0-255)
    analogWrite(pwmB, speed);
    delay(2000);

    // === Stop ===
    analogWrite(pwmA, 0);
    analogWrite(pwmB, 0);
    delay(1000);

    // === Reverse ===
    digitalWrite(dirA1, LOW);
    digitalWrite(dirA2, HIGH);

```

```
digitalWrite(dirB1, LOW);  
digitalWrite(dirB2, HIGH);  
  
analogWrite(pwmA, speed);  
analogWrite(pwmB, speed);  
delay(2000);  
  
// === Stop again ===  
analogWrite(pwmA, 0);  
analogWrite(pwmB, 0);  
delay(1000);  
}
```

Controlling 2 DC motors simultaneously

Discussion

The experimental results demonstrate that the L298P Motor Driver Shield effectively controls both the speed and direction of a DC motor using Pulse Width Modulation (PWM) and GPIO logic signals. The observed data align closely with theoretical expectations, confirming that the PWM duty cycle directly influences the average voltage applied to the motor, which in turn determines its rotational speed. The proportional relationship between PWM and speed validates the fundamental principle of PWM-based motor control, where higher duty cycles produce faster rotations due to increased average power delivery.

The directional control, managed through the IN1 and IN2 input pins, also performed as anticipated. The motor rotated forward when IN1 was HIGH and IN2 was LOW, and reversed direction when the inputs were swapped. When both inputs were set HIGH, the motor stopped immediately, indicating active braking — a behaviour consistent with the expected operation of an H-bridge configuration. This confirmed the L298P's capability to provide bidirectional motor control efficiently.

Despite the overall accuracy of the results, minor discrepancies were observed between theoretical and estimated rotational speeds. These deviations can be attributed to several factors:

1. Voltage fluctuations from the external power supply, which may have caused slight variations in actual motor voltage.
2. Mechanical friction and motor wear, which can reduce rotational speed compared to the ideal theoretical model.
3. Human estimation error in RPM measurement, as the speed was estimated visually rather than using a tachometer.
4. PWM signal resolution limits, where the 8-bit resolution (0–255) of Arduino PWM may not perfectly represent linear voltage increments, introducing small non-linearities in speed response.

These limitations suggest that the precision of the experiment could be improved by employing a digital tachometer or an optical encoder to obtain more accurate RPM measurements. Additionally, using an oscilloscope to verify the PWM waveform could ensure that the signal maintains the expected duty cycle under varying loads.

The implications of these findings are significant in the context of mechatronic system design. The experiment confirms that PWM control via the L298P driver is an efficient and practical method for managing DC motor performance. Such control systems are essential in modern automation applications, including robotic actuation, conveyor control, and

servo-based mechanisms, where precision, energy efficiency, and bidirectional motion are critical.

Overall, the discussion highlights that while minor measurement uncertainties existed, the experiment successfully verified the theoretical principles of PWM-based motor speed control and H-bridge direction control. The results strongly support the use of the L298P Motor Driver Shield as a reliable component for embedded motor control systems.

Conclusion

The experiment was done successfully and all the objectives were met. The L298P Motor Driver Shield operating principle was clearly comprehended, as well as how they regulate the direction and speed of DC motors with the help of the GPIO pins. By adoption of the PWM (Pulse Width Modulation), the motor speed was accurately regulated by changing the duty cycle, exhibiting the association amongst the voltage and the motor speed. To test the proper integration of hardware and software, the Arduino IDE made it possible to write, upload, and run the program successfully. On the whole, the experiment was a good practical experience of operating the DC motors on the L298P driver and PWM.

Recommendations

- Use a tachometer to measure the motor revolutions per minute (RPM) accurately instead of using slow motioned video to estimate it.
- Use an external power source to provide stable operation.
- Add a current sensor to monitor the power consumption of the motor so that efficiency under different loads can be studied.
- Apply a potentiometer to control variable speed smoothly.
- Never power the motor directly from Arduino's 5V pin as the motors are inductive loads that can create voltage spikes which will harm the Arduino board.
- Understanding how the duty cycle affects motor speed will provide clear insights on how varying PWM speed controls the speed of the motor.

References

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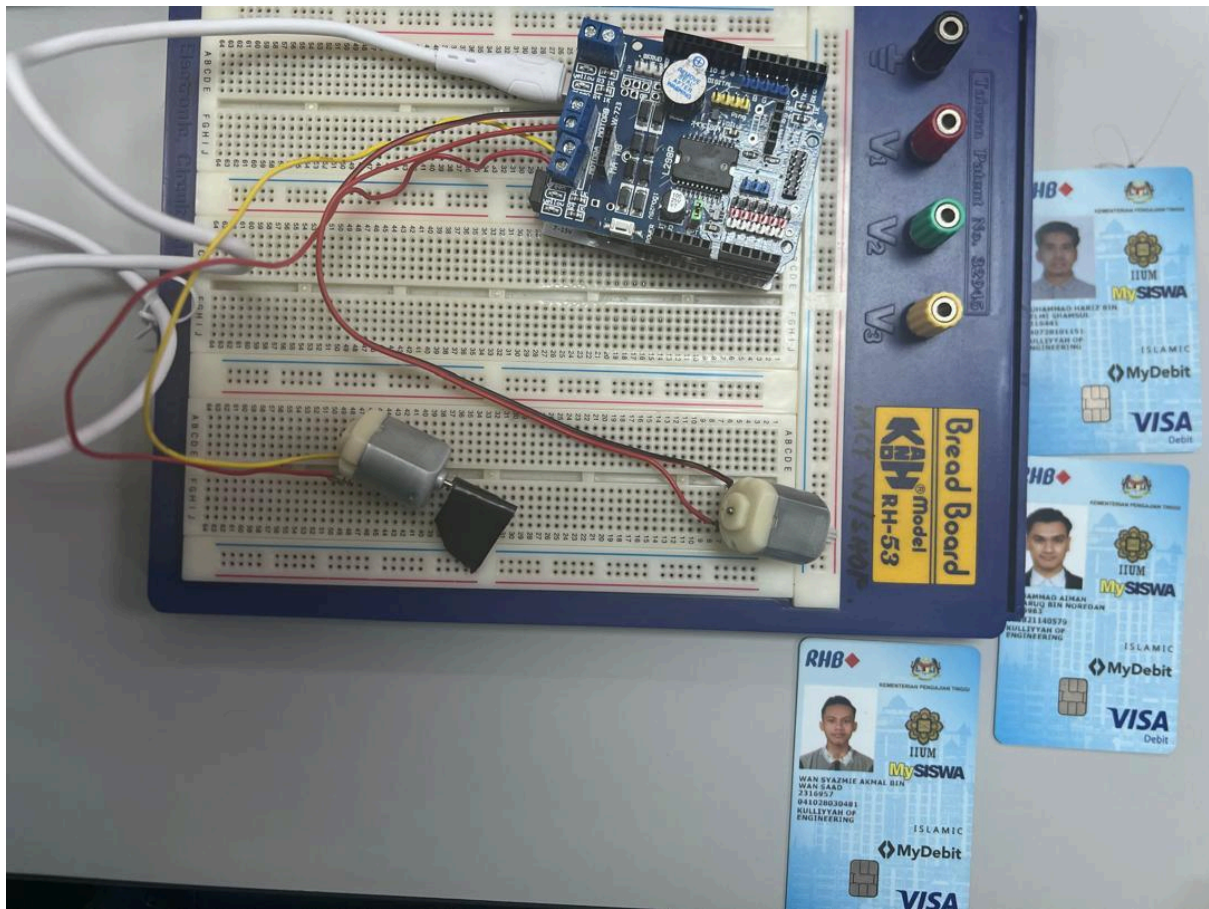
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Appendices



Circuit Design

Shield Pin	Arduino Pin	Function
ENA	D3	PWM (Speed A)
IN1	D12	Motor A Direction 1
IN2	D13	Motor A Direction 2
ENB	D5	PWM (Speed B)
IN3	D10	Motor B Direction 1
IN4	D11	Motor B Direction 2

Pin reference (Arduino UNO)

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Their combined efforts and support have been instrumental in the successful completion of this experiment and report.

Student's Declaration

This is to certify that we are responsible for the work submitted in this report, that the original work is our own except as specified in the references and acknowledgement, and that the original work contained herein have not been undertaken or done by unspecified sources or persons.

We hereby certify that this report has not been done by only one individual and all of us have contributed to the report. The length of contribution to the reports by each individual is noted within this certificate.

We also hereby certify that we have read and understand the content of the total report and no further improvement on the reports is needed from any of the individual's contributors to the report.

We therefore, agreed unanimously that this report shall be submitted for marking and this final printed report has been verified by us.

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